One-class hybrid SVM-CAE based MEWMA monitoring scheme

Stephen Pienaar

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Supervisor(s): Dr. MJC Malela, Co-supervisor(s): Dr. Lyson Chaka

Department of Statistics, University of Pretoria



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Abstract

This study develops and evaluates a hybrid monitoring scheme for nonlinear, serially dependent multivariate processes that combines a one-dimensional convolutional autoencoder (CAE), a one-class support vector machine (OC-SVM) and a multivariate exponentially weighted moving average (MEWMA) control chart. For every window of 32 observations, the CAE's reconstruction error and the OC-SVM's anomaly score on the CAE features form a two-dimensional monitoring vector that the MEWMA accumulates over time. The scheme is studied on a simulated nonlinear process driven by a latent vector autoregression, using five independent datasets for representation learning, detector fitting, empirical calibration of the control limit, independent in-control (IC) validation, and out-of-control (OOC) evaluation. The results reveal that the hybrid chart has a shorter OOC average run-length (ARL) than both of its single-component ablations at every shift, but it does not outperform the classical MEWMA: the two are indistinguishable at the smallest shift, and the classical chart detects faster at every shift of 0.4 or more. Post-hoc diagnostics attribute this mainly to the weak response of distance-type learned features when the mean shifts in a fixed direction, amplified at large shifts by the startup behaviour of the charts. Simulated Extended Tennessee Eastman Process data were used to demonstrate the application of the proposed method. The frozen design was refitted to 31 process variables and both charts were calibrated to a 5% probability of at least one false alarm over 568 nominal decisions instead of an IC ARL of 370. The classical MEWMA again detected the feed-composition fault IDV(2) sooner at every severity, by 4.8 to 10.1 monitoring decisions on average in onset-referenced conditional detection delay, while the hybrid chart's median delay was 17 to 39 decisions (0.80 to 1.90 hours after onset), depending on severity.

Keywords: Statistical process control; MEWMA chart; Convolutional autoencoder; One-class support vector machine; Multivariate process monitoring; Anomaly detection; Tennessee Eastman Process.

Declaration

I, *Stephen James Pienaar,* declare that this research report, submitted in partial fulfillment of the degree *BCom(Hons) Statistics and Data Science,* at the University of Pretoria, is my own work and has not been previously submitted at this or any other tertiary institution.

Artificial intelligence (AI) use disclosure

Artificial intelligence tools, including advanced language models such as ChatGPT and Claude/Claude Code, were used in the preparation of this report to support grammatical clarity, formatting and document structure, literature-search assistance, comprehension of the research topic, and code debugging. All research questions, methodological decisions, data analysis, interpretation of findings, and final conclusions were critically reviewed by the author and remain the author's own responsibility. No generative AI output was accepted without author review and verification.

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Stephen James Pienaar

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Supervisor: Dr. MJC Malela

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Co-supervisor: Dr. Lyson Chaka

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Date

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1 Introduction

1.1 General concepts in SPC and ML

1.1.1 Statistical Process Control

Statistical process control (SPC) is a collection of statistical techniques for monitoring process stability and improving capability by reducing variability [16]. Its principal tool is the control chart, which plots a monitoring statistic against a control limit; a value beyond the limit signals a possible out-of-control (OOC) departure from the in-control (IC) state. Charts are judged by their run-lengths, usually through the average run-length (ARL): a long IC ARL (ARL_0) means few false alarms and a short OOC ARL (ARL_1) fast detection [21]. Memoryless charts use only the current observation, which makes them responsive to large shifts but insensitive to small ones; the Shewhart chart, for example, signals when the current sample mean falls outside three-sigma lower and upper control limits (LCL and UCL) around the IC mean [21]. Memory-type charts such as the cumulative sum (CUSUM) [18] and the exponentially weighted moving average (EWMA) [24] accumulate evidence and suit small, persistent shifts; the multivariate EWMA (MEWMA) [12] extends this to correlated characteristics. These charts assume independent, normally distributed observations, and when either assumption fails the attained ARL_0 can differ substantially from its nominal value [12, 21]. These limitations motivate more flexible monitoring schemes, including nonparametric [2] and machine learning (ML)-based control charts [17].

1.1.2 Machine Learning

ML algorithms learn patterns from training data and so offer alternative ways to identify process shifts, faults and anomalies without a parametric process model; as labelled faults are rare, relevant methods model only normal behaviour [19]. An autoencoder is a neural network trained to reconstruct its input through a low-dimensional bottleneck [7]. Once it is trained on IC data, large reconstruction errors signal departures such as anomalies or shifts. A convolutional autoencoder (CAE) extracts local patterns from short multivariate time-series windows [9, 15]. The one-class support vector machine (OC-SVM), an unsupervised extension of the support vector machine [4], encloses most normal data in a boundary and scores new observations against it [25]. On CAE features the two signals provide complementary information: the score indicates whether a window's learned representation is typical of normal process behaviour, whereas the reconstruction error reflects how accurately the network reproduces the window.

1.2 Problem statement and significance of the study

Learned detectors can represent the nonlinear, serially dependent structure that weakens standard control limits, and they are usually applied observation by observation against a fixed threshold, without accu-

mulating evidence or a run-length interpretation of their alarm rate [3, 6, 13]. This study places CAE reconstruction information and OC-SVM anomaly-score information inside a memory-type MEWMA chart and evaluates the result as a control chart: its limit is calibrated to a nominal ARL_0 and its run-lengths are compared with those of a classical MEWMA calibrated to the same target, so any benefit of learning is judged on the classical chart's own terms.

1.3 Research aim and objectives

The aim of this study is to develop and evaluate a hybrid ML-based MEWMA monitoring scheme that integrates a CAE and an OC-SVM for detecting process shifts in nonlinear multivariate process data. The specific objectives of the study are to:

- construct a CAE and OC-SVM hybrid scheme that combines both signals in one monitoring vector;
- investigate its run-length characteristics, with particular interest in small shifts;
- evaluate the CAE and OC-SVM components using appropriate diagnostics;
- compare the hybrid chart with a classical MEWMA under a common IC design target;
- examine its performance across shift magnitudes and three smoothing constants; and
- demonstrate the proposed hybrid framework on simulated data, including an industrial benchmark.

1.4 Research outline

The study is confined to unsupervised monitoring of simulated processes, so the true state is always known: a nonlinear process built for the primary experiment and, as an external application, the Extended Tennessee Eastman Process (TEP) benchmark. It covers the Phase I regime, in which all components and limits are estimated from IC data, and the Phase II regime, in which the frozen scheme is applied to independent sequences [21]. Section 2 reviews the literature on control charts and ML for process monitoring. Section 3 describes the experimental design, data generation, control charts, ML models and evaluation criteria. Section 4 presents the primary simulation results, Section 5 applies the approach to the TEP, and Section 6 summarises the findings, recommendations, limitations and future work.

2 Literature Review

For monitoring multiple correlated characteristics, two principal monitoring approaches have been developed. The first extends univariate charts through Hotelling's T^2 statistic and, for the memory-type case, culminates in the MEWMA chart. Its limit is set to achieve a specified ARL_0, and its ARL depends on a

mean shift only through the shift's Mahalanobis distance [12]. The second, reviewed by MacGregor and Kourti [14], projects the observations onto a few latent variables and monitors two complementary statistics: T^2 on the retained scores and a squared prediction error (SPE) on the residual. Both approaches rely on linear structures and the assumptions of independence and normality of the observations.

Representation learning offers a nonlinear counterpart to the projection step. Autoencoders learn low-dimensional codes that outperform principal components for dimension reduction [7], and one-dimensional convolutional networks suit sensor signals when labelled data are scarce [8]. The analogy with latent-variable monitoring is direct: the autoencoder plays the role of the principal component model and its reconstruction error that of the SPE. For the latent space, feeding learned codes to an OC-SVM is established practice [1, 13, 20], and Guzman and Howe [6] report that combining autoencoder reconstruction error with an OC-SVM beats either component alone.

Hybrid ML and SPC research has followed two strands. In the first, a learner operates on charting statistics while the chart remains classical: convolutional and recurrent networks recognise control chart patterns [26], deep networks and support vector regression (SVR) model EWMA-type statistics for censored reliability data [10, 11], and models trained on classical charting statistics widen the range of detectable shifts [17]. These learners need labelled OOC examples or a parametric model from which training data are simulated. In the second strand, representation learning enters the monitoring statistic itself: Biegel et al. [3] compute T^2 and SPE statistics from a deep autoencoder's latent representation and reconstruction error on a real sheet-metal forming process. That approach is unsupervised and acts on process observations, but its statistics are memoryless and it is assessed by classification metrics rather than run-lengths. Within the literature reviewed, no study combines a CAE reconstruction error with an OC-SVM score on the CAE representation in a memory-type chart whose limit is calibrated to a nominal ARL_0 and compared with a classical MEWMA under the same target; this study addresses that gap.

3 Methodology

3.1 Simulated process and study design

A latent state $\boldsymbol{z}_t \in \mathbb{R}^8$ follows a stable mean-adjusted VAR(1), and the four observed variables $\boldsymbol{x}_t \in \mathbb{R}^4$ are a nonlinear function of it:

$$\begin{aligned} \boldsymbol{z}_t &= \boldsymbol{\mu}_t + \boldsymbol{\Phi}(\boldsymbol{z}_{t-1} - \boldsymbol{\mu}_{t-1}) + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim N(\boldsymbol{0}, 0.2\boldsymbol{I}_8), \\ \boldsymbol{x}_t &= \boldsymbol{W}_2 \tanh(\boldsymbol{z}_t) + \boldsymbol{\varepsilon}_t \quad \text{and} \quad \boldsymbol{\varepsilon}_t \sim N(\boldsymbol{0}, 0.05\boldsymbol{I}_4), \end{aligned} \tag{1}$$

where $\boldsymbol{\eta}_t$ and $\boldsymbol{\varepsilon}_t$ are independent Gaussian latent innovations and measurement noise, $\boldsymbol{I}_k$ is the $k \times k$ identity, $\boldsymbol{\mu}_t = \boldsymbol{0}$ in the IC state, $\boldsymbol{\Phi}$ is symmetric tridiagonal (diagonal 0.55, adjacent off-diagonals 0.15),

tanh acts componentwise and $\boldsymbol{W}_2$ is a fixed 4×8 mixing matrix (Appendix A). The pre-specified entries of $\boldsymbol{\Phi}$ give each latent component persistence and neighbouring components dependence while keeping the process stationary (spectral radius $\approx 0.832 < 1$); as $0.832^{32} \approx 0.003$, a 32-observation window spans almost all of this persistence. Each trajectory starts at $\boldsymbol{z}_0 = \boldsymbol{0}$ and discards 500 burn-in steps. Because the map is nonlinear and $p = 4 < 8$, no linear projection of $\boldsymbol{x}_t$ recovers $\boldsymbol{z}_t$; the latent state is never supplied to any model. The fault is a sustained latent mean shift $\boldsymbol{\mu}_t = \delta \boldsymbol{v}$ in the fixed direction $\boldsymbol{v} = (1, 1, 0, \ldots, 0)^\top / \sqrt{2}$, starting at the first retained observation ($\tau = 1$); δ is the Euclidean size of the shift. Fifteen magnitudes, $\delta = 0.2, 0.4, \ldots$ and 3.0, are studied; no other fault type is used.

Five datasets generated from independent random streams have separate roles (Figure 1): In Phase I, D1 trains the CAE, D2 fits the OC-SVM and D3 estimates the chart moments and calibrates the limits; everything is then frozen and applied unchanged to D4 (independent IC validation) and D5 (OOC evaluation) in Phase II.

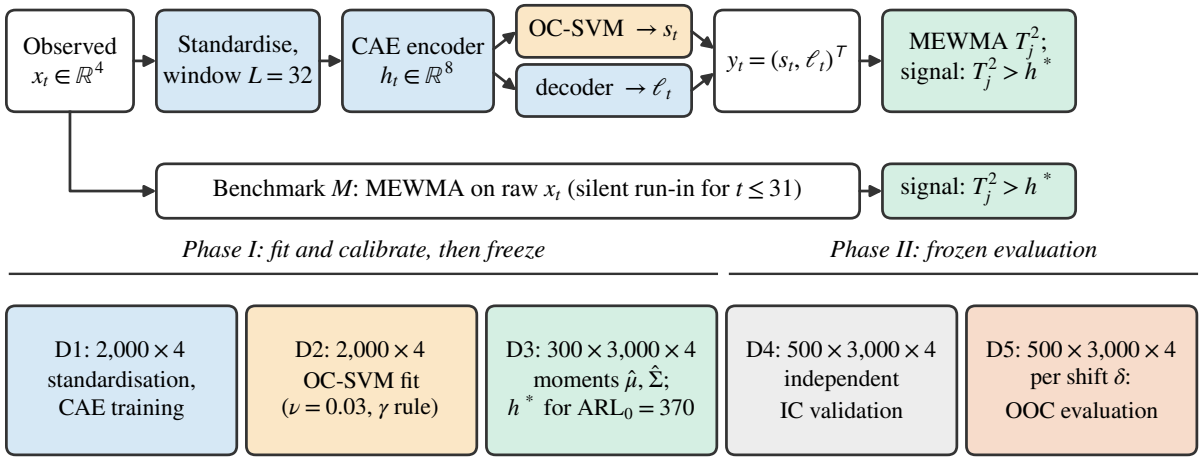


Figure 1: Monitoring paths of the hybrid chart H (ablations: S uses s_t only, R uses ℓ_t only) and the benchmark M (top), and the roles of the five datasets (bottom). Box shading marks the dataset on which a stage is fitted or calibrated.

3.2 Learned components

Windows: D1 is split chronologically into training rows $[0, 1600)$ and validation rows $[1600, 2000)$ before any window is formed. Each variable is standardised with the training-segment mean and standard deviation, which are then frozen. Windows $\boldsymbol{X}_t \in \mathbb{R}^{4 \times 32}$ of the $L = 32$ most recent standardised observations are formed with stride 1 inside a segment or trajectory, so a 3000-observation trajectory yields $J = 2969$ windows, the first ending at $t = 32$.

CAE: The encoder applies two one-dimensional convolutions ($4 \rightarrow 8 \rightarrow 16$ channels), which halve the window length twice, and a linear bottleneck producing $\boldsymbol{h}_t \in \mathbb{R}^q$ with $q = 8$; the decoder mirrors it to give the reconstruction $\hat{\boldsymbol{X}}_t$ (3180 parameters; layers and optimiser settings in Appendix A). The network

was trained on the 1569 D1 training windows to minimise the mean squared reconstruction error, and the checkpoint with the lowest validation loss (epoch 33) was frozen. The reconstruction error is the summed squared prediction error of the standardised window,

$$\mathrm{SPE}_t = \sum_{c=1}^{4} \sum_{k=1}^{32} \big(X_{t,ck} - \hat{X}_{t,ck}\big)^2, \tag{2}$$

where c indexes channels and k window positions. The charts use $\ell_t = \log \max(\mathrm{SPE}_t, 10^{-12})$: the floor prevents $\log 0$, and the logarithm puts the error on a scale comparable to the OC-SVM score and removes most of its right skew, without making it Gaussian.

OC-SVM: The 1969 D2 windows are passed through the frozen standardisation and encoder, and an RBF OC-SVM is fitted to the resulting features $\boldsymbol{h}_i$ directly, without further scaling:

$$k(\boldsymbol{h}, \boldsymbol{h}') = \exp\big(-\gamma \|\boldsymbol{h} - \boldsymbol{h}'\|^2\big), \qquad \nu = 0.03 \quad \text{and} \quad \gamma = \Big[\operatorname*{median}_{i<j} \|\boldsymbol{h}_i - \boldsymbol{h}_j\|^2\Big]^{-1} \approx 0.0305, \tag{3}$$

with the median taken over all $\binom{1969}{2}$ pairs. The parameter ν bounds the fraction of training windows outside the fitted boundary from above and the fraction of support vectors from below [25]. The score is the negated decision function, $s_t = \rho - \sum_{m=1}^{78} \alpha_m k(\boldsymbol{h}_{(m)}, \boldsymbol{h}_t)$, where $\boldsymbol{h}_{(1)}, \ldots, \boldsymbol{h}_{(78)}$ are the support vectors, α_m their dual coefficients and ρ the fitted offset, so a larger s_t means a more abnormal window.

3.3 Monitoring charts and calibration

The hybrid monitoring vector is $\boldsymbol{y}_t = (s_t, \ell_t)^\mathsf{T}$. For a chart input $\boldsymbol{u}_j$ with IC mean $\hat{\boldsymbol{\mu}}$ and covariance $\hat{\boldsymbol{\Sigma}}$, the MEWMA state and the plotting statistic are, respectively, [12]:

$$\boldsymbol{Z}_j = \lambda \boldsymbol{u}_j + (1-\lambda)\boldsymbol{Z}_{j-1}, \qquad \boldsymbol{Z}_0 = \hat{\boldsymbol{\mu}}, \tag{4}$$

$$T_j^2 = (\boldsymbol{Z}_j - \hat{\boldsymbol{\mu}})^\mathsf{T} (c_j \hat{\boldsymbol{\Sigma}})^{-1} (\boldsymbol{Z}_j - \hat{\boldsymbol{\mu}}), \tag{5}$$

where j indexes monitoring opportunities ($j = 1$ at $t = 32$), $\boldsymbol{u}_j$ is the chart's input at opportunity j (for H, $\boldsymbol{u}_j = \boldsymbol{y}_{j+31}$), λ is the smoothing constant and $c_j = \frac{\lambda}{2-\lambda}\big[1 - (1-\lambda)^{2j}\big]$ is the exact finite-start factor, which is always used. Six configurations of four charts are evaluated (Table 1). The hybrid chart (denoted as $H(\lambda)$), an OC-SVM–CAE MEWMA chart, monitors $\boldsymbol{y}_t$ at the primary $\lambda = 0.05$, with $\lambda = 0.10$ and 0.20 as a sensitivity analysis; the ablations $S(\lambda)$ and $R(\lambda)$ monitor s_t and ℓ_t alone, and the benchmark $M(0.05)$ is a conventional MEWMA on the raw $\boldsymbol{x}_t$, each with $\lambda = 0.05$. $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$ are pooled over all 300×2969 D3 monitoring vectors (S and R use the matching entries) and, for M, over the aligned raw D3 observations at $t = 32, \ldots, 3000$. Because the hybrid charts cannot decide before the first window is complete, M's state is updated silently with $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{31}$ and it first reports at $t = 32$ with factor c_{32}, so

all six charts share the same 2969 opportunities.

Consecutive windows share 31 of their 32 observations, so the monitored sequences are strongly autocorrelated (Section 4.1) and $c_j\hat{\boldsymbol{\Sigma}}$ is only a reference scaling, not the covariance of $\boldsymbol{Z}_j$; without independence, limits derived for independent data lose their nominal ARL properties [12]. Each limit is therefore calibrated empirically on D3. For a candidate limit h, the run-length (RL) of a trajectory is the first opportunity j with $T_j^2 > h$, truncated at $J = 2969$, and $\widehat{ARL}_0(h)$ is the mean RL over the 300 D3 trajectories. As this is a non-decreasing step function of h, the calibrated limit

$$h^* = \min\left\{h \in V : \widehat{ARL}_0(h) \geq 370\right\}, \tag{6}$$

where V is the set of values attained by the running maxima of T_j^2, is found exactly by bisection, and the chart signals at the first j with $T_j^2 > h^*$, where $h^* > 0$ is its UCL. The target of 370 is the ARL_0 of a conventional three-sigma Shewhart chart [21].

3.4 Evaluation design

If a chart gives no signal by $j = 2969$, the RL is recorded as 2969 and flagged as censored by a separate indicator, so each mean is a restricted ARL. D4 yields each chart's independent IC run-lengths, and D5 its OOC run-lengths at the 15 shifts. Because $\tau = 1$, the first window is already post-change and a signal at opportunity j corresponds to $j + 31$ post-change observations: ARL_1 is a detection delay measured from the first permitted decision, the setting in which Lowry et al. [12] also compared multivariate charts. $H(0.05)$ and $M(0.05)$ monitor the same D5 trajectories, so the primary comparison is paired,

$$\bar{d}(\delta) = \frac{1}{500}\sum_{i=1}^{500}\left[RL_{H,i}(\delta) - RL_{M,i}(\delta)\right], \tag{7}$$

with standard error $s_d/\sqrt{500}$, where $RL_{H,i}(\delta)$ is the RL of $H(0.05)$ on D5 trajectory i at shift δ (similarly for M) and s_d the sample standard deviation of the 500 paired differences; a positive $\bar{d}$ means that the benchmark signalled first. The design, including every dataset size, was fixed in advance: no model, moment, smoothing constant or limit was changed after D4 or D5 was examined.

4 Results

4.1 Behaviour of the learned components

The CAE generalised to independent IC data, with a mean squared reconstruction error per window element of 0.66 on the D1 validation windows and 0.67 on D3, so the diagnostics do not indicate substantial overfitting to its 1569 training windows. The OC-SVM met its ν-property in sample (3.0% of D2 windows

outside the boundary; 4.0% support vectors), but 6.2% of D3 windows fell outside, so its boundary tightened somewhat out of sample, which the empirical chart calibration absorbs. Figures 2(a)–(b) show that the components were weakly correlated ($r = -0.15$), that s_t was right-skewed (skewness 0.55) while ℓ_t was nearly symmetric, and that both were strongly serially dependent: their lag-one autocorrelation of 0.96 decayed almost linearly to near zero at lag 32, where windows stop sharing observations, whereas the raw variables' lag-one autocorrelations of 0.45–0.60 fell below 0.07 by lag 10.

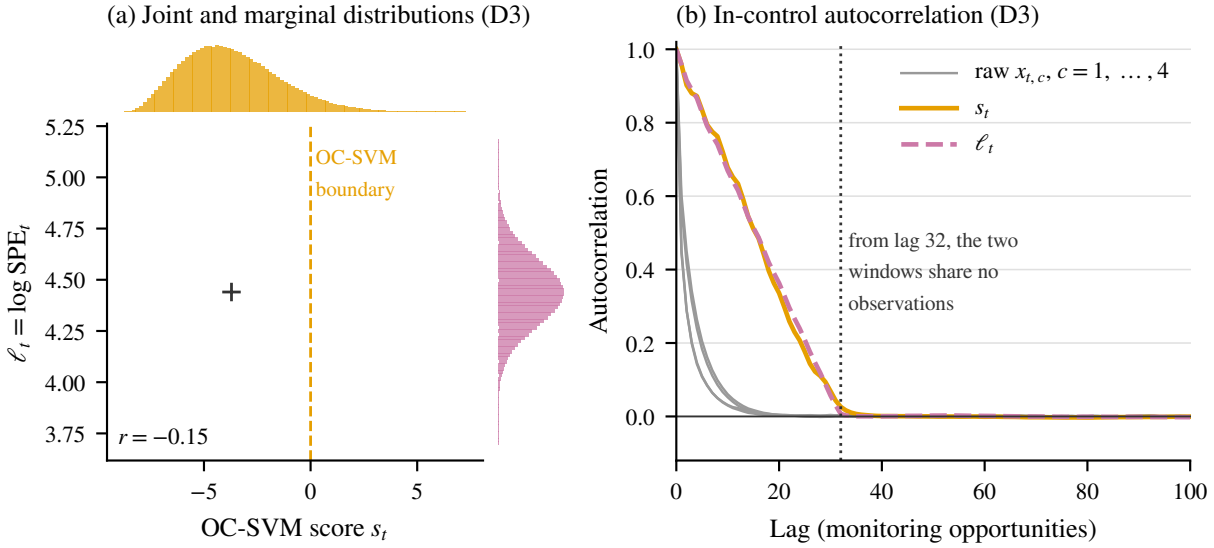


Figure 2: IC monitoring vectors on D3 (300×2969 windows). (a) Joint distribution of (s_t, ℓ_t) (grey shading: log count; cross: $\hat{\boldsymbol{\mu}}$) with marginal histograms. (b) Autocorrelation of s_t, ℓ_t and the four raw variables, pooled over trajectories.

4.2 IC validation

Table 1 sets each chart's D3 calibration value beside its independent D4 performance; no D4 run-length was censored (one D3 trajectory was censored when calibrating $H(0.20)$). The attained D4 ARL_0 departed from 370 in opposite directions: the hybrid configurations ran shorter (2.3–4.0 D4 SE below target), the ablations stayed close and $M(0.05)$ ran longer (1.9 SE above); on the same D4 trajectories, $M(0.05)$'s run-lengths exceeded those of $H(0.05)$ by 67.0 on average (paired SE 22.2). Deviations of this size can arise from sampling variability in both the finite calibration bank and D4, but their direction matters: $H(0.05)$ was the more liberal chart in the IC state, which, if anything, favours it in the OOC state.

4.3 Primary comparison: $H(0.05)$ versus $M(0.05)$

Figures 3(a)–(b) cover the complete shift grid; no D5 run-length was censored. At $\delta = 0.2$ the charts were indistinguishable ($ARL_1 = 282.4$ and 285.6; $\bar{d} = -3.2$, paired SE 15.9). At every $\delta \geq 0.4$ the benchmark, although the more conservative chart in the IC state, signalled first on average, by more than ten paired

standard errors (Appendix C). The absolute gap peaked at $\delta = 0.4$–0.6 ($\bar{d} \approx 98$–106 opportunities) and then narrowed, while the relative gap widened: $M(0.05)$ signalled at its first opportunity in 41% of trajectories at $\delta = 1.0$ and in essentially all from $\delta = 2.2$, whereas $H(0.05)$ never did so and levelled off near an ARL_1 of 12.5. Counted in post-change observations (j+31), the hybrid chart needed 1.4–2.5 times as many as the benchmark for $\delta \geq 0.4$, against up to about 22 times as many monitoring opportunities.

Table 1: Calibrated limit, D3 calibration ARL_0 and independent D4 IC performance (500 trajectories; SE in parentheses) of the six charts.

Chart	Input	h^*	D3 ARL_0	D4 ARL_0 (SE)	D4 *SDRL*	D4 *MRL*
$H(0.05)$	$\boldsymbol{y}_t = (s_t, \ell_t)^\top$	111.08	371.0	336.5 (14.7)	328.8	230.5
$H(0.10)$	$\boldsymbol{y}_t = (s_t, \ell_t)^\top$	83.59	370.5	322.3 (13.6)	304.2	226.0
$H(0.20)$	$\boldsymbol{y}_t = (s_t, \ell_t)^\top$	51.43	372.4	315.4 (13.7)	305.9	210.5
$S(0.05)$	s_t	66.05	370.6	351.4 (14.9)	333.0	246.5
$R(0.05)$	ℓ_t	71.03	371.3	363.4 (16.1)	361.0	253.5
$M(0.05)$	raw $\boldsymbol{x}_t$	46.52	372.4	403.5 (17.9)	401.3	268.5

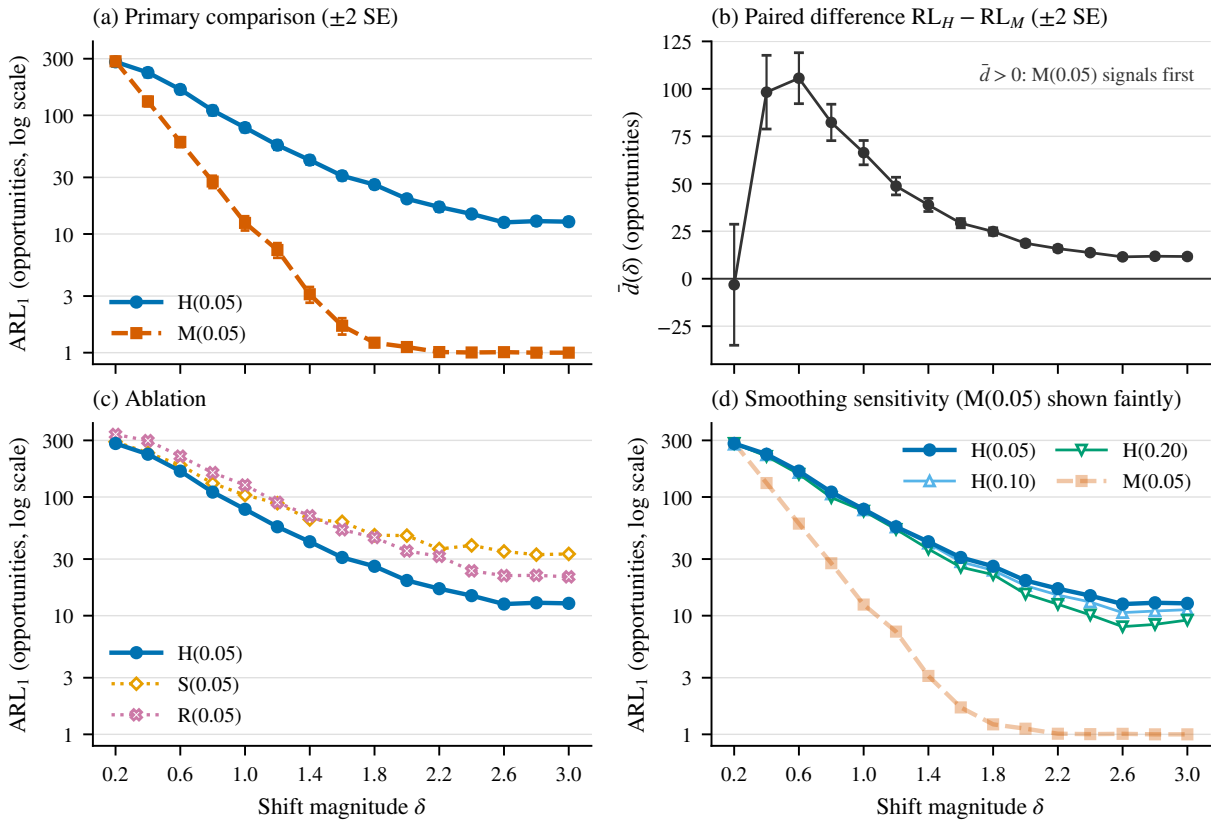


Figure 3: OOC results on D5 (500 trajectories per δ; log scale in (a), (c), (d)). (a) ARL_1 of $H(0.05)$ and $M(0.05)$; (b) paired mean difference $\bar{d}(\delta)$ of Equation (7); (c) $H(0.05)$ and its ablations; (d) the three hybrid smoothing constants, with $M(0.05)$ for reference.

4.4 Ablation and smoothing sensitivity

Figure 3(c) shows that $H(0.05)$ had a lower ARL_1 than both ablations at all 15 shifts. Paired over the same trajectories (Appendix C), its advantage over $R(0.05)$ was clear throughout (8.6–69.0 opportunities, more than three paired SE), whereas that over $S(0.05)$ was within sampling error at $\delta \leq 0.4$ and exceeded two paired SE from $\delta = 0.6$; the score was the stronger single component up to $\delta = 1.4$ and the reconstruction error beyond it. Figure 3(d) shows that a larger λ changed little at small shifts and helped at larger ones, as expected for EWMA-type charts [12]: $H(0.20)$ had the lowest hybrid ARL_1 at every $\delta \geq 0.4$, by 2.6–11.0 opportunities relative to $H(0.05)$. Its D4 ARL_0 was also lower (paired difference -21.1, SE 11.0), so part of this gain may reflect its more liberal IC behaviour, and no smoothing constant in the pre-specified range brought the hybrid chart close to $M(0.05)$.

4.5 Interpretation

Two features of the design explain these results. First, the fault is a sustained mean shift in a fixed direction, the alternative that a linear, accumulating MEWMA on $\boldsymbol{x}_t$ is built to detect [12], whereas the hybrid chart accumulates two distance-type summaries of each window, which respond weakly to small location shifts. Second, with $\tau = 1$ the benchmark's first report already contains 32 accumulated post-change updates, whereas the hybrid chart's first statistic is the Mahalanobis distance of a single window, compared with a limit set by the strongly persistent steady state. Post-hoc diagnostics, which are not part of the pre-specified experiment, support both explanations: the component means moved by only 0.10–0.21 IC standard deviations at $\delta = 0.4$ (Appendix B), and a separate diagnostic audit estimated that $\boldsymbol{y}_t$ retains 2–36% of the raw data's mean-shift information per update, that startup mechanics account for most of the gap in monitoring opportunities at $\delta \geq 1.8$, and that with the change after 100 IC reports $H(0.05)$ still needed 2.0–2.5 times as many post-change observations as $M(0.05)$ for $\delta \geq 0.6$. These diagnostics explain the canonical result; they do not change it.

5 Application

This section applies the frozen design to an external benchmark: the Tennessee Eastman Process (TEP) of Downs and Vogel [5], a simulated closed-loop chemical plant with a reactor, separator and stripper under feedback control, in the extended version released by Reinartz et al. [22] as a fault-detection benchmark. Its faults have known onsets and severities, so detection can be timed against the truth, and its purpose here is to test whether the frozen design transfers to a larger closed-loop process. It is a simulation, not measured plant data, and the synthetic study remains the primary experiment.

5.1 Benchmark and frozen protocol

In Mode 1, the data are sampled every 3 minutes and are nominal until the fault activates after 30 hours [23]. The fault is the process disturbance IDV(2): a step in the B composition of feed stream 4, with the ratio of components A and C (A/C) held constant [22], at severities of 25, 50, 75 and 100%. The 31 variables ($p = 31$) under consideration are the 22 continuous measurements and the nine non-constant manipulated variables. Whole runs are allocated to roles: nominal pre-fault segments of other runs form TEP-D1 to TEP-D4, in the roles of D1–D4, and TEP-D5 comprises all 500 IDV(2) runs (Appendix E).

$H(0.05)$ and $M(0.05)$ follow the guidelines of Sections 3.2–3.3, with $L = 32$, $q = 8$, $\nu = 0.03$ and $\lambda = 0.05$. Only the CAE's channel count changed, and every component was refitted on TEP Phase I data and frozen before TEP-D4 and TEP-D5 were read, without tuning. The dataset contains only five independent long nominal runs, so an ARL_0 of 370 cannot be estimated. Each limit instead gives a 5% probability of at least one false alarm over the 568 nominal decisions, and no ARL equivalent is claimed.

To evaluate the delay in fault detection, each chart runs from the start of the simulation without restarts, and a run is eligible for a chart if that chart gave no alarm before onset. The onset-referenced conditional detection delay D counts decisions from onset to the first signal ($D = 1$ at 30.00 h; $0.05(D-1)$ hours elapsed; censoring at $D = 1401$). H and M are compared on jointly eligible runs with the paired estimator of Equation (7); a positive $D_H - D_M$ means that the hybrid chart was slower. Because D follows IC operation and conditions on no earlier alarm, it is not an ARL_1, and severity is not comparable with δ. The hybrid chart's window at delay D holds only the smaller of D and 32 post-change observations, whereas M sees the change from $D = 1$.

5.2 Results and interpretation

On TEP-D4 the false-alarm probability was 0.0435 for $H(0.05)$ and 0.0425 for $M(0.05)$ (exact 95% intervals 0.035–0.053 and 0.034–0.052), and before onset in TEP-D5 it was 0.036 and 0.038. Both charts therefore met the criterion and, unlike in Section 4.2, were matched in the IC state. Delays fell monotonically with severity (Figure 4; medians in Table 2), and $M(0.05)$ was faster at every severity. The mean paired differences $D_H - D_M$ ranged from +10.1 to +4.8 decisions (about 0.51 to 0.24 h), each many standard errors above zero, and the hybrid chart was slower in every jointly eligible run at 25–75% and in 185 of 188 at 100%.

Part of the gap is structural: at 50–100% every eligible hybrid signal came before its window was entirely post-change ($D < 32$), and only at 25% did its median delay (39) come later. The criterion bounds false alarms only over the 568 nominal decisions, although all delays (7–49 decisions) fell well within the 1401-decision horizon. The 100% cell is confounded with a separate seed batch, which showed no consistent IC batch effect, and only IDV(2) was evaluated (see Figure 4).

Table 2: Extended TEP, IDV(2): onset-referenced conditional detection delay D in monitoring decisions of 3 min (elapsed hours $0.05(D-1)$ in parentheses) and the numbers of runs eligible for $H(0.05)$, $M(0.05)$ and both. Eligibility requires no pre-onset alarm; a positive $D_H - D_M$ means that the hybrid chart was slower. No eligible delay was censored.

		Eligible	Median D (hours)		Paired mean
Severity	Runs	$H/M/$paired	$H(0.05)$	$M(0.05)$	$D_H - D_M$ (SE)
25%	100	96/98/96	39 (1.90)	29 (1.40)	+10.1 (0.4)
50%	100	94/96/91	25 (1.20)	18 (0.85)	+6.9 (0.2)
75%	100	96/97/96	20 (0.95)	14 (0.65)	+5.5 (0.2)
100%	200	196/190/188	17 (0.80)	12 (0.55)	+4.8 (0.1)

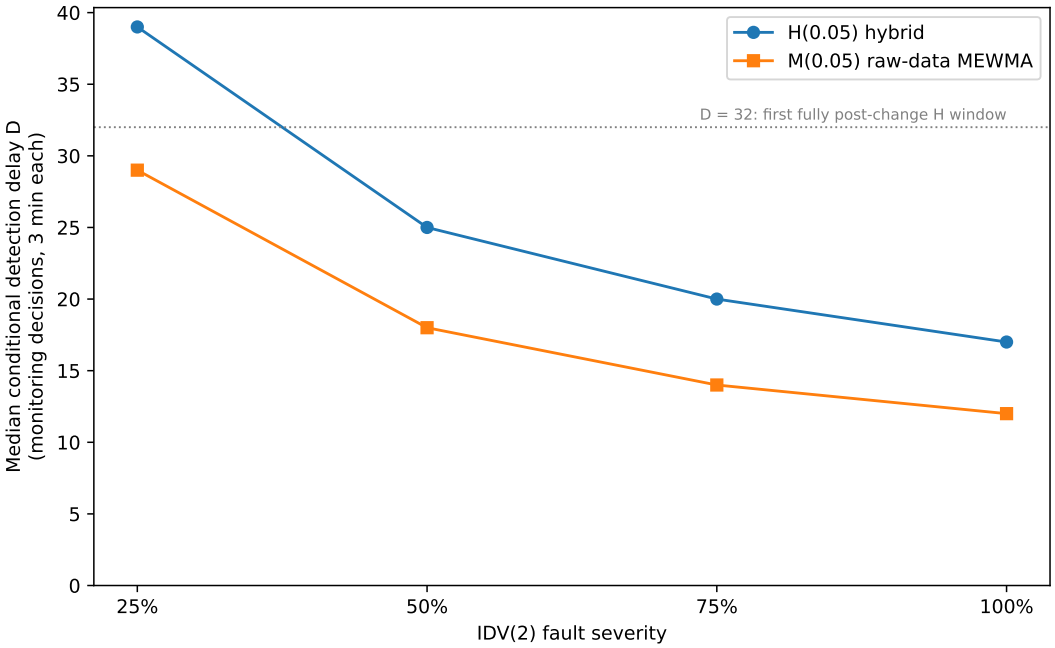


Figure 4: Extended TEP, IDV(2): onset-referenced conditional detection delay. Median onset-referenced conditional detection delay of $H(0.05)$ and $M(0.05)$ against IDV(2) severity (eligible runs of Table 2); the dotted line marks $D = 32$, the first entirely post-change hybrid window.

The frozen framework thus transfers, calibrates to a horizon-appropriate criterion and responds coherently to the fault, but shows no advantage over the classical chart. IDV(2) is a persistent step disturbance, which a raw-data MEWMA accumulates efficiently [12], and the ordering matches the primary comparison. Because the onset followed 568 IC decisions, the startup mechanism of Section 4.5 cannot explain this gap, although the ramp-in can explain part of it. Deployment would also need a genuinely IC reference period and refitting after legitimate process changes, and its signals are harder to attribute to variables than those of a raw-data MEWMA, whose state vector shows the shift direction [12].

6 Conclusion

6.1 Summary

This study developed a hybrid scheme in which a MEWMA chart accumulates a CAE reconstruction error and an OC-SVM score of the CAE features. In the primary experiment, on a simulated nonlinear process with five independent datasets, every chart was calibrated empirically to an IC ARL of 370. The learned components behaved largely as intended, and on independent IC data the hybrid chart was somewhat more liberal, and the classical benchmark somewhat more conservative, than the target. The hybrid chart had a lower ARL_1 than both single-component ablations at every shift (within sampling error only against $S(0.05)$ at $\delta \leq 0.4$), so the combined vector carries more usable information than either learned signal alone, but it did not outperform the classical MEWMA: the two were indistinguishable at $\delta = 0.2$, and the benchmark was faster at every larger shift, with the hybrid chart needing 1.4 to 2.5 times as many post-change observations. Larger smoothing constants shortened the hybrid chart's run-lengths modestly without closing the gap. In the external Extended TEP benchmark, with limits set by a false-alarm probability rather than an ARL, both charts met their IC target and the classical MEWMA detected IDV(2) sooner at every severity, by 4.8 to 10.1 decisions on average.

6.2 Recommendations

First, for sustained mean shifts in a moderately nonlinear process and persistent step disturbances such as TEP IDV(2), a well-calibrated classical MEWMA on the raw variables should remain the default, and a learned hybrid scheme should be preferred only where it demonstrably improves detection of the relevant disturbance structure under a common IC criterion; because both studies are simulations, this implication has yet to be confirmed on measured plant data. Second, learned reconstruction and one-class information should be monitored jointly, since $H(0.05)$ had lower mean ARL_1 than either ablation at every evaluated shift. Third, charts on overlapping windows should be calibrated empirically on the dependent process, with delays reported both in monitoring opportunities and in post-change observations.

6.3 Future work

The hybrid framework should be evaluated on faults that alter the shape, variability or dependence of the data (variance changes, sensor faults, drifts and transients), including the other TEP faults, for which distance-type features may be better suited, and on other processes. Direction-preserving learned features, dependence-aware finite-start scaling for overlapping windows, windowing that shortens the post-change ramp-in, and validation on measured plant data also deserve study.

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Appendix A: Simulation and model details

Table 3 gives the fixed mixing matrix of Equation (1) (full row rank; the weights of each observed variable have unit norm) and Table 4 the layers of the convolutional autoencoder, both as implemented in the authoritative repository. The CAE was trained with Adam (learning rate 10^{-3}, batch size 64, at most 100 epochs, early-stopping patience 10).

Table 3: The fixed 4×8 mixing matrix $\boldsymbol{W}_2$, shown transposed: row k holds the weights of $\tanh(z_{t,k})$ in the four observed variables $x_{t,1}, \ldots, x_{t,4}$.

k	$x_{t,1}$	$x_{t,2}$	$x_{t,3}$	$x_{t,4}$
1	−0.60447198	−0.04842081	−0.05518327	−0.04100004
2	−0.06150649	0.36085142	0.45874929	−0.84305316
3	−0.37179578	−0.83557846	−0.21939336	−0.40588165
4	−0.01161746	−0.11135410	−0.17469105	−0.11509976
5	0.05885824	−0.18152197	−0.55464952	0.26627250
6	0.52361117	0.30422580	0.30578469	0.07542546
7	0.45120458	0.15832824	−0.13652085	0.16952036
8	−0.10604239	0.07924137	−0.53672635	−0.06530930

Table 4: Layers of the convolutional autoencoder (kernel 3, stride 2 and padding 1 for every convolution; output padding 1 for the transposed convolutions).

Stage	Layer	Output shape	Parameters
Input	standardised window $\boldsymbol{X}_t$	4×32	–
Encoder	Conv1d $4 \to 8$, ReLU	8×16	104
	Conv1d $8 \to 16$, ReLU	16×8	400
	flatten, Linear $128 \to 8$ (bottleneck $\boldsymbol{h}_t$)	8	1032
Decoder	Linear $8 \to 128$, ReLU; unflatten	16×8	1152
	ConvTranspose1d $16 \to 8$, ReLU	8×16	392
	ConvTranspose1d $8 \to 4$ (reconstruction $\hat{\boldsymbol{X}}_t$)	4×32	100
Total			3180

Appendix B: Supplementary component diagnostics

Figure 5 shows a typical IC window and its frozen-CAE reconstruction: the decoder reproduces the smooth level structure of each channel but not the high-frequency variation, which therefore dominates SPE_t. Figure 6 compares the IC distributions of the standardised components with their distributions under three shifts. It is a post-hoc descriptive diagnostic computed from the first 100 D5 trajectories at each shift with the frozen models; it was not a pre-specified output of the experiment.

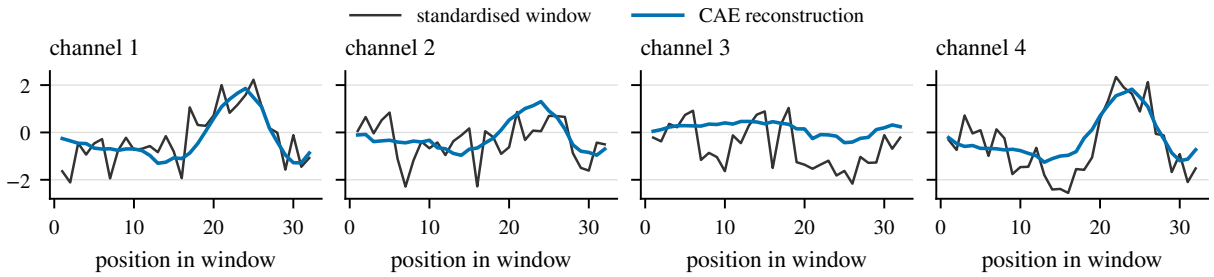


Figure 5: One IC D3 window (trajectory 1, window 1001) and its reconstruction by the frozen CAE ($\mathrm{SPE}_t = 96.7$, against a D3 mean of 86).

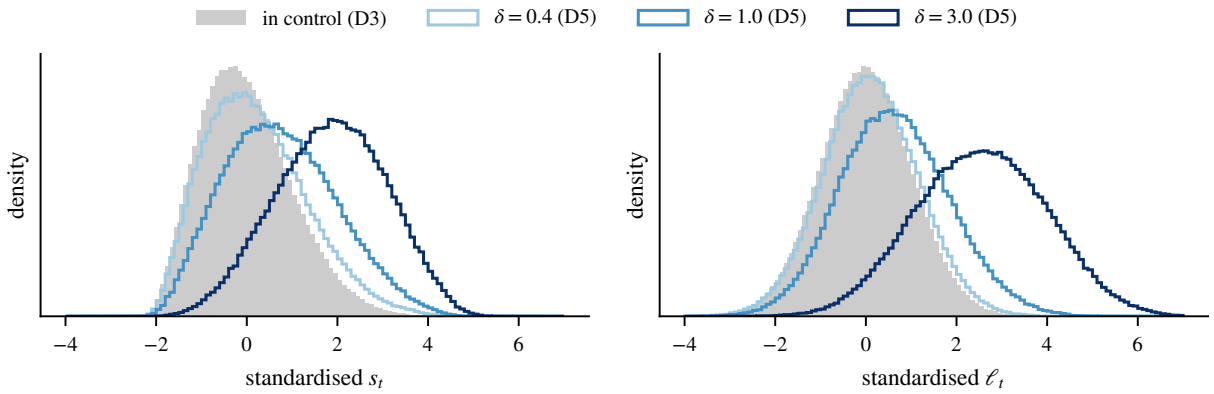


Figure 6: Post-hoc descriptive diagnostic: distributions of s_t and ℓ_t, standardised by the D3 moments, IC (D3) and at $\delta = 0.4$, 1.0 and 3.0 (first 100 D5 trajectories each). Mean shifts: s_t 0.21, 0.75, 1.86 and ℓ_t 0.10, 0.64, 2.55 standard deviations.

Appendix C: Supplementary Phase II results

Figure 7 is the repository's own summary of the D4 validation in Table 1. Tables 5–7 give the complete OOC results; every entry is computed from the committed per-trajectory run-lengths, and no D5 run-length was censored. Figure 8 shows run-length distributions of the primary pair at four shifts.

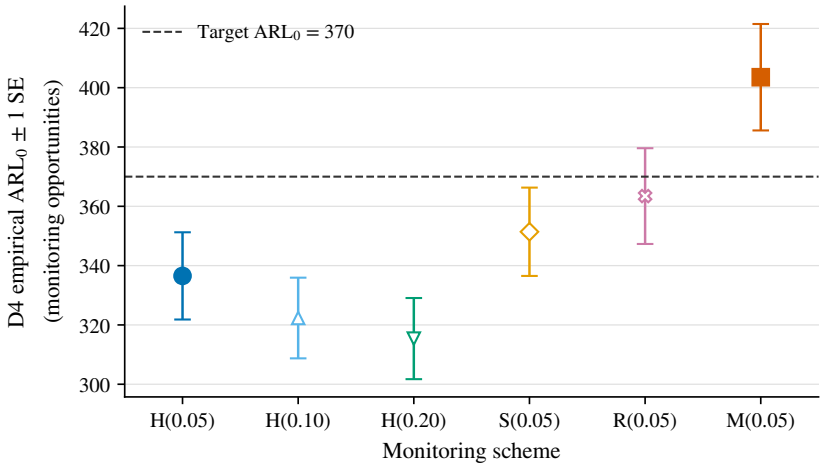


Figure 7: D4 IC *ARL* (±1 SE, 500 trajectories) of the six charts against the target of 370 (repository figure).

Table 5: Restricted OOC *ARL* (SE) of all six charts across the shift grid (D5, 500 trajectories per shift).

δ	$H(0.05)$	$H(0.10)$	$H(0.20)$	$S(0.05)$	$R(0.05)$	$M(0.05)$
0.2	282.4 (12.0)	278.1 (11.9)	283.6 (12.9)	292.1 (13.4)	336.6 (14.7)	285.6 (12.7)
0.4	229.4 (9.8)	226.3 (9.9)	220.5 (9.9)	235.7 (9.9)	298.4 (13.8)	131.1 (6.0)
0.6	165.3 (6.8)	160.5 (7.0)	155.8 (7.1)	185.8 (8.1)	220.7 (11.3)	59.6 (2.9)
0.8	110.0 (4.9)	105.8 (4.9)	99.0 (4.9)	130.4 (5.4)	160.5 (6.6)	27.7 (1.6)
1.0	78.8 (3.3)	77.7 (3.4)	76.2 (3.6)	103.8 (4.4)	126.1 (5.6)	12.4 (0.9)
1.2	56.1 (2.4)	55.8 (2.6)	53.5 (2.6)	88.6 (3.7)	90.1 (3.9)	7.3 (0.5)
1.4	42.0 (1.8)	40.8 (1.9)	36.4 (1.8)	64.7 (2.9)	69.4 (2.9)	3.1 (0.2)
1.6	30.9 (1.2)	28.4 (1.3)	25.6 (1.3)	61.8 (2.8)	53.0 (2.3)	1.7 (0.1)
1.8	26.1 (1.0)	24.1 (1.1)	22.2 (1.1)	47.7 (2.1)	45.6 (2.0)	1.2 (0.0)
2.0	19.8 (0.7)	17.9 (0.7)	15.2 (0.8)	47.3 (2.0)	35.1 (1.7)	1.1 (0.0)
2.2	16.9 (0.7)	14.9 (0.7)	12.5 (0.7)	36.5 (1.7)	31.7 (1.4)	1.0 (0.0)
2.4	14.7 (0.5)	13.0 (0.5)	10.1 (0.5)	39.1 (1.7)	23.9 (1.1)	1.0 (0.0)
2.6	12.5 (0.4)	10.6 (0.4)	8.1 (0.4)	34.8 (1.7)	21.8 (1.0)	1.0 (0.0)
2.8	12.8 (0.5)	10.9 (0.5)	8.4 (0.5)	32.7 (1.5)	21.9 (1.1)	1.0 (0.0)
3.0	12.7 (0.5)	11.2 (0.5)	9.2 (0.5)	33.1 (1.4)	21.3 (1.0)	1.0 (0.0)

Table 6: Primary comparison in detail: paired mean difference $\bar{d}$ (SE), median run-lengths, share of $M(0.05)$ trajectories signalling at the first opportunity (no $H(0.05)$ trajectory did at any shift), mean delay in post-change observations ($ARL_1 + 31$) and the ratio of the two charts' delays in both units.

		MRL			Obs. delay		Ratio H/M	
δ	$\bar{d}$ (SE)	H	M	$P(RL_M = 1)$	H	M	obs.	opp.
0.2	−3.2 (15.9)	205.0	190.0	0.024	313.4	316.6	0.99	1.0
0.4	98.2 (9.7)	164.0	89.5	0.068	260.4	162.1	1.61	1.7
0.6	105.6 (6.7)	117.0	41.5	0.122	196.3	90.6	2.17	2.8
0.8	82.3 (4.8)	79.0	16.0	0.242	141.0	58.7	2.40	4.0
1.0	66.4 (3.2)	54.5	4.5	0.408	109.8	43.4	2.53	6.4
1.2	48.8 (2.3)	37.5	1.0	0.572	87.1	38.3	2.27	7.6
1.4	38.9 (1.7)	29.0	1.0	0.744	73.0	34.1	2.14	13.5
1.6	29.2 (1.2)	21.0	1.0	0.902	61.9	32.7	1.89	18.3
1.8	24.9 (1.0)	18.0	1.0	0.944	57.1	32.2	1.77	21.5
2.0	18.7 (0.7)	14.0	1.0	0.982	50.8	32.1	1.58	17.7
2.2	15.9 (0.7)	12.0	1.0	0.998	47.9	32.0	1.50	16.7
2.4	13.7 (0.5)	11.0	1.0	0.998	45.7	32.0	1.43	14.7
2.6	11.5 (0.4)	10.0	1.0	0.998	43.5	32.0	1.36	12.4
2.8	11.8 (0.5)	9.0	1.0	1.000	43.8	32.0	1.37	12.8
3.0	11.7 (0.5)	9.0	1.0	1.000	43.7	32.0	1.37	12.7

Table 7: Paired mean run-length differences (SE) of $H(0.05)$ against its ablations; negative values mean that $H(0.05)$ signalled sooner.

δ	$RL_H - RL_S$	$RL_H - RL_R$
0.2	−9.7 (14.5)	−54.1 (15.5)
0.4	−6.4 (11.3)	−69.0 (14.2)
0.6	−20.6 (8.1)	−55.4 (10.6)
0.8	−20.4 (5.3)	−50.5 (6.6)
1.0	−25.0 (4.2)	−47.3 (5.1)
1.2	−32.5 (3.5)	−33.9 (3.7)
1.4	−22.7 (2.8)	−27.4 (2.7)
1.6	−30.9 (2.7)	−22.1 (2.2)
1.8	−21.6 (1.9)	−19.5 (1.8)
2.0	−27.5 (2.0)	−15.3 (1.6)
2.2	−19.6 (1.6)	−14.8 (1.3)
2.4	−24.3 (1.7)	−9.1 (1.0)
2.6	−22.2 (1.6)	−9.2 (0.9)
2.8	−19.8 (1.4)	−9.0 (1.0)
3.0	−20.4 (1.4)	−8.6 (0.9)

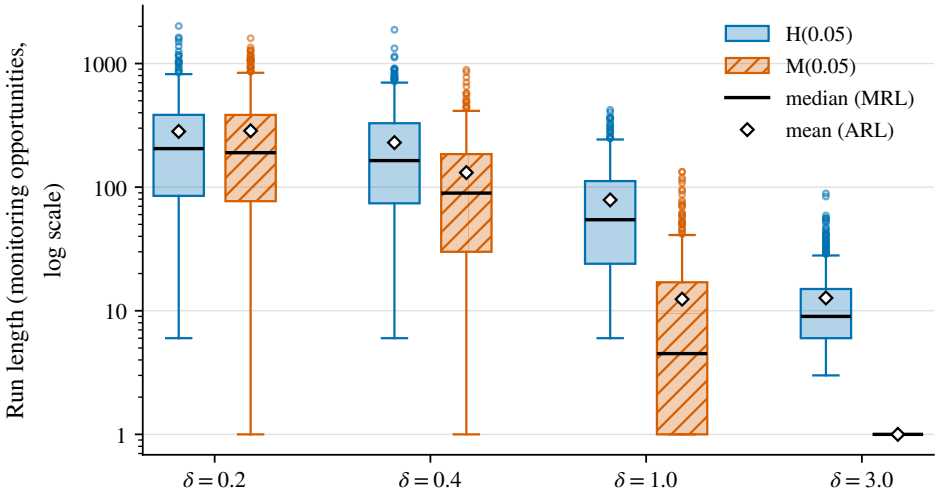


Figure 8: Run-length distributions of $H(0.05)$ and $M(0.05)$ at four shifts (log scale; repository figure).

Appendix D: Reproducibility and provenance

The primary experiment is the committed state of `pipeline-thesis-independent`, the authoritative repository (branch `simplify-pipeline-code`, commit `e62bdee`). Data generation uses master seed 795 with independent PCG64DXSM streams per dataset, trajectory and shift; the CAE training seed is 3795; all numerical routes run in double precision (network in single precision) on one CPU thread. The frozen preprocessing statistics, CAE weights, OC-SVM state, D3 moments and control limits, and the per-trajectory D4/D5 run-lengths are committed artifacts, and the production run that produced them took 207 s. Every synthetic figure and generated table in this report is rebuilt, without refitting or recalibration, by `scripts/make_all_figures.py` in the report workspace, which reads these artifacts read-only; the source and derivation of each figure are recorded in `planning/final_figure_provenance.md`. The post-hoc diagnostic audit cited in Section 4.5 is a separate analysis of the same frozen experiment and is not part of its pre-specified design. The TEP application has its own provenance, given in Appendix E.

Appendix E: Extended TEP protocol and provenance

The TEP application is a separate frozen study with its own protocol. It is the committed state of the repository `pipeline-thesis-tep` (branch `tep-application`, commit `0a121fa`), read in this workspace through `08_tep_application/source`. The data are the Mode 1 file `TEP_Mode1.h5` of the published dataset [23], whose MD5 checksum (`c79ecc6d...`) matches the official record. The protocol and the run registry were frozen before any model was fitted (SHA-256 `558ffcb6...` and `0c03350d...`).

Data and roles: Rows before 30.00 h are IC, so each run has 599 nominal rows and 568 pre-onset decisions, and each IDV(2) run has 2000 rows and 1969 decisions. TEP-D1 to TEP-D4 were drawn, with generator seed $(795, 6)$, from the 13,445 non-IDV(2) runs whose seed occurs only once in the dataset: TEP-D1 has 40 training and 10 validation runs, TEP-D2 20 runs, and TEP-D3 and TEP-D4 2000 runs each. The monitored variables are XMEAS(1)–XMEAS(22) and XMV(1)–XMV(4), XMV(6)–XMV(8), XMV(10) and XMV(11). XMV(5), XMV(9) and XMV(12) were constant on the TEP-D1 training rows and were removed, and the composition analysers XMEAS(23)–XMEAS(41) and the additional measurements were never used.

Phase I: The CAE (seed 3795) reached its lowest validation loss, a mean squared reconstruction error of 0.677 in standardised units, at epoch 52. The OC-SVM, fitted to 11,360 TEP-D2 windows, has $\gamma = 0.0208$ and 361 support vectors. The raw covariance for M had condition number 4.05×10^8, above the pre-registered guard of 10^8, so M was computed in the frozen standardised coordinates (condition number 280), which leaves T_j^2 unchanged; its 31-observation silent run-in is as in Section 3.3. Each limit is the 1900th smallest of the 2000 TEP-D3 per-run maxima of T_j^2 over the 568 nominal decisions: 325.46

for $H(0.05)$ and 423.59 for $M(0.05)$, each exceeded by exactly 100 maxima. Both charts start at $\boldsymbol{Z}_0 = \hat{\boldsymbol{\mu}}$ at the first observation of every run and are never restarted. Phase I ran in about 80 s on one processor thread.

Validation and evaluation: On TEP-D4, 87 of the 2000 runs gave a false alarm for $H(0.05)$ and 85 for $M(0.05)$. The false-alarm probability was 0.0375 for $H(0.05)$ and 0.0442 for $M(0.05)$ in the 1200 runs from the dataset's 25–75% seed-generation batch, and 0.0525 and 0.0400 in the 800 runs from its separate 100% batch; these values are descriptive only and changed nothing. Before onset in TEP-D5, $H(0.05)$ signalled in 4, 6, 4 and 4 runs and $M(0.05)$ in 2, 4, 3 and 10 runs at the four severities. Between 91 and 196 runs were eligible per cell. The standard deviations of D were 4.4, 2.2, 1.7 and 1.7 decisions for $H(0.05)$ and 3.8, 2.1, 1.5 and 1.4 for $M(0.05)$, and the paired standard errors of $D_H - D_M$ were 0.42, 0.24, 0.19 and 0.12.

Presentation files: Figure 4 is the frozen TEP figure redrawn by its own plotting code from the frozen summary, with only the in-plot title removed at the supervisor's request; its plotted values are unchanged. The rows of Table 2 are taken from the frozen primary table and summary (`d5_primary_table.csv` and `d5_summary.json`). No result was recomputed, and no model was refitted, recalibrated or re-evaluated for this report. Source paths and SHA-256 values are recorded in `planning/final_figure_provenance.md`.